

Vehicle service Booking and Tracking System

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PROBLEM STATEMENT

- Vehicle service booking is often handled through calls or manual processes.
- Customers may not know the current status of their vehicle.
- Service centers need efficient management of bookings and job cards.
- Spare parts, repairs, billing, and service history may be maintained separately.
- Customers need better communication and transparency during servicing.
- Manual processes can lead to delays, errors, and difficulty in maintaining records.

OBJECTIVES

- Simplify the vehicle service booking process.
- Allow customers to manage multiple vehicles.
- Provide service-center selection based on preferred location.
- Provide service slot selection and booking management.
- Track vehicle service progress through different stages.
- Manage job cards, technicians, inspections, and spare parts.
- Maintain invoices, payments, and service history.
- Provide maintenance reminders and service notifications.

PROPOSED SYSTEM

- A centralized system for managing the complete vehicle servicing process.
- Customers can register vehicles and book services.
- Customers can select a preferred service center and available time slot.
- Service centers can create and manage digital job cards.
- Technicians can be assigned based on service requirements.
- Vehicle inspection and additional repair approval can be managed digitally.
- Spare parts, billing, invoices, and payments can be maintained systematically.
- Completed services are stored as service history for future reference.

Literature suvey

Aspect	Existing Systems	Our Proposed Project
Booking	Basic online appointment booking	Online booking with service, center and time-slot selection
Service Tracking	Limited or basic status updates	Detailed vehicle service status tracking
Job Cards	May be manual or limited	Digital job-card management
Technician Management	Limited technician allocation	Technician allocation and workload management
Vehicle Inspection	Often handled manually	Digital inspection records
Additional Repairs	Customer approval may be handled manually	Digital repair request with customer approval/rejection

Advantages and Disadvantages

Advantages of Our Project

- Centralizes the complete vehicle service lifecycle.
- Improves transparency through service tracking and repair approval.
- Reduces manual work using digital job cards, inspections and inventory.
- Provides separate interfaces for customers and service centers.
- Maintains organized billing, payment and service-history records.

Disadvantages / Limitations

- Requires internet connectivity for online access.
- Backend development and integration are more complex because of multiple modules.
- Requires proper data security and access control.
- Real-time notifications, maps and payment services may require third-party APIs.
- Initial implementation and maintenance may require additional time and resources.

MODULES

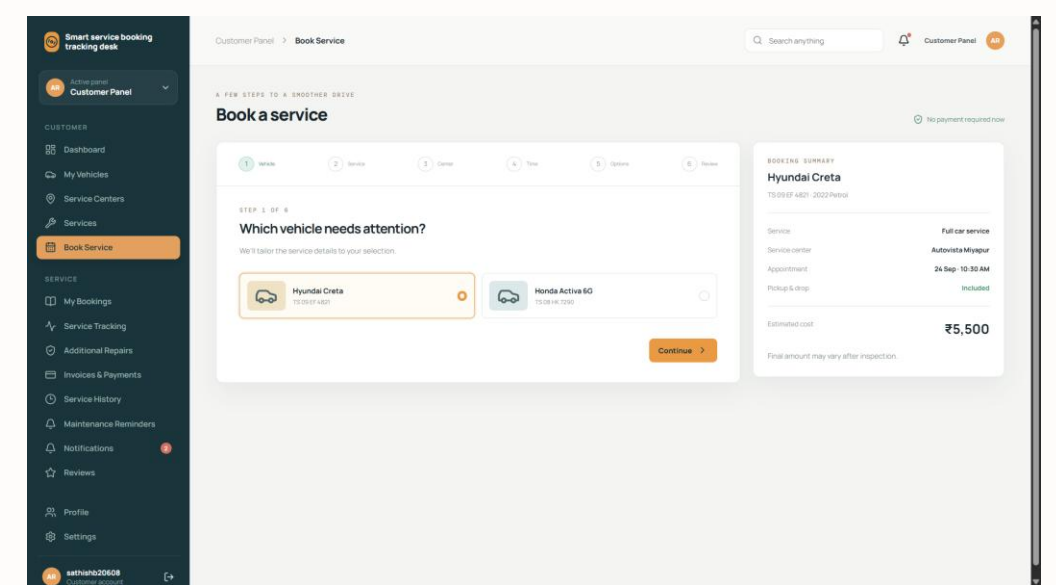
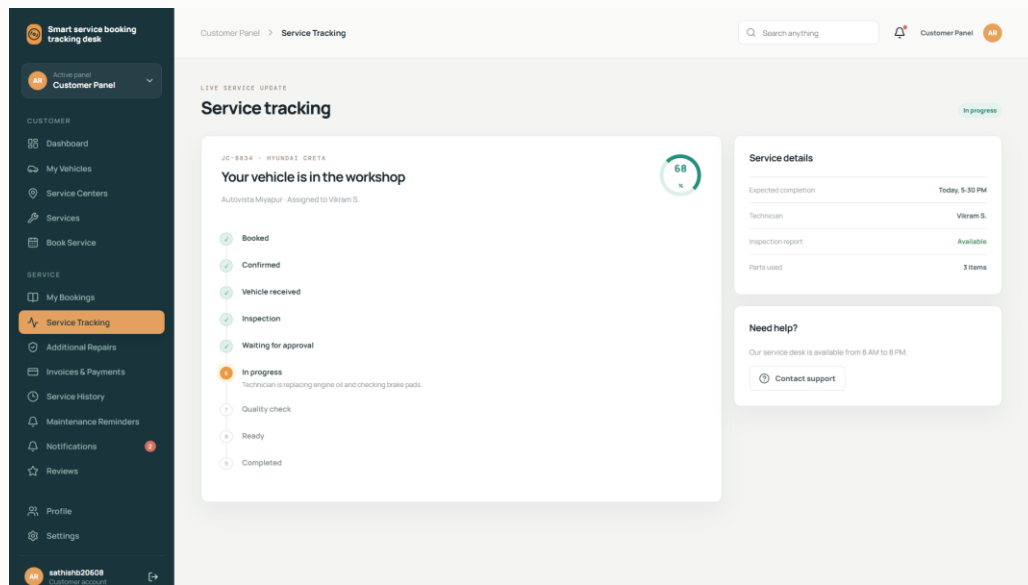
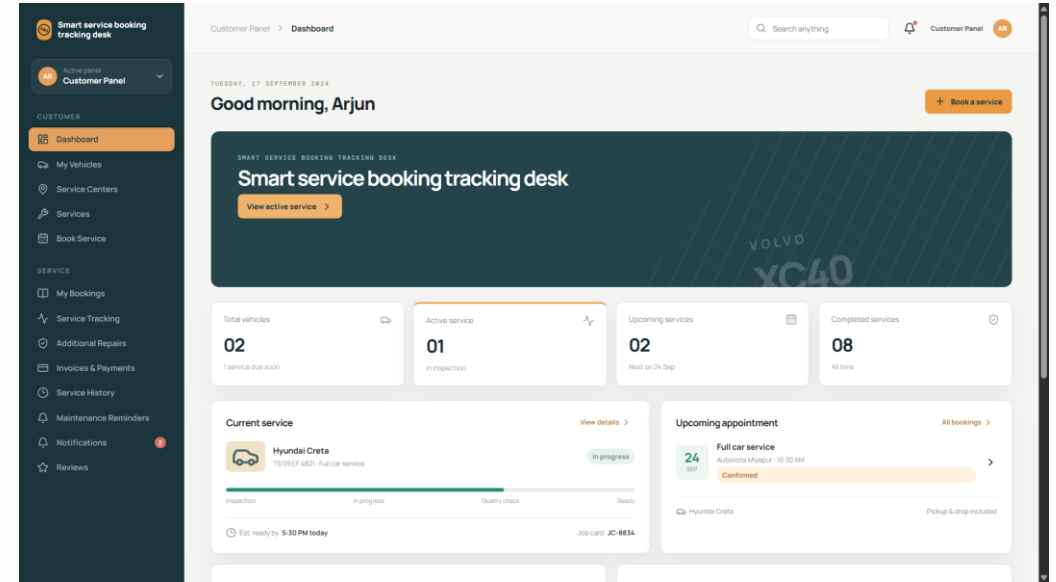
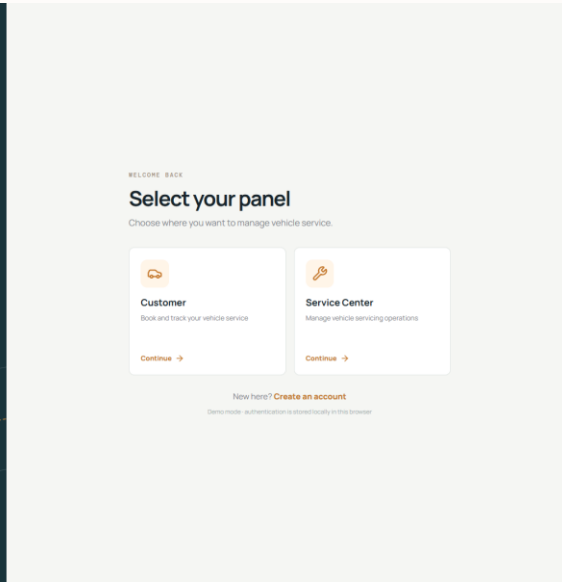
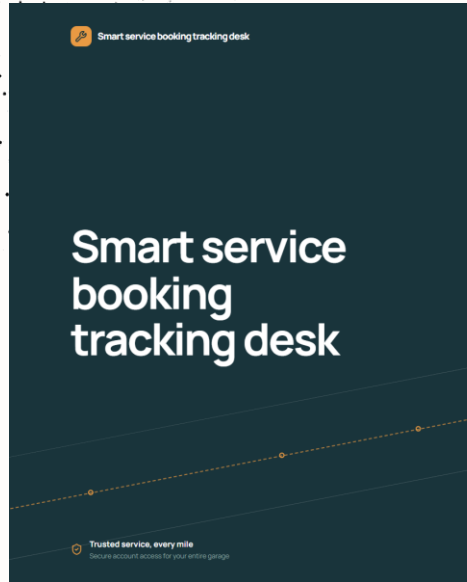
Customer Modules

- Dashboard
- My Vehicles
- Service Centers
- Services
- Book Service
- My Bookings
- Service Tracking
- Additional Repairs
- Invoices & Payments
- Service History
- Maintenance Reminders
- Notifications
- Reviews

Service Center Modules

- Booking Management
- Job Cards
- Technician Allocation
- Vehicle Inspection
- Spare Parts
- Inventory
- Additional Repairs
- Billing & Invoices
- Customer Management

Frontend Implementation – Screenshots

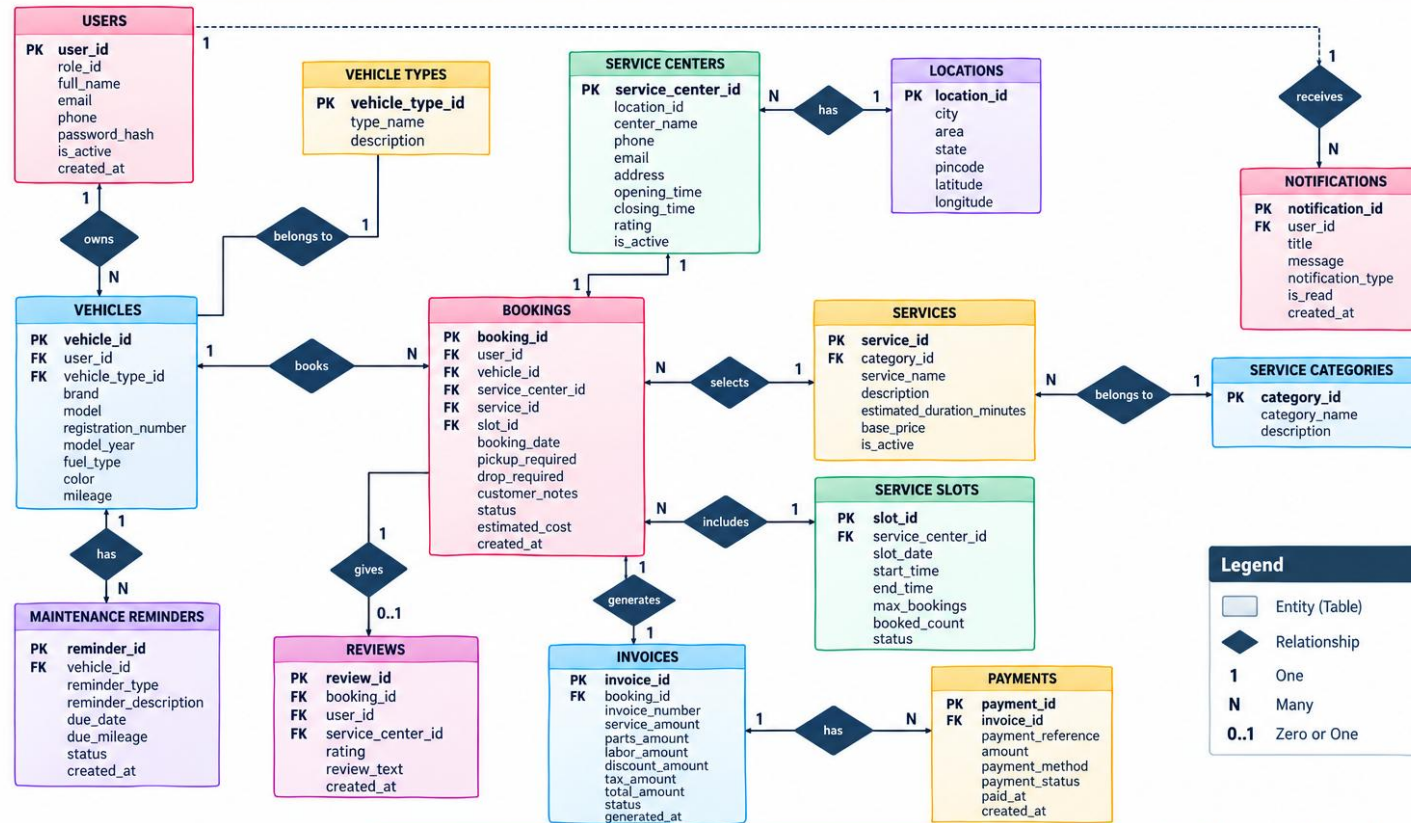


ER Diagram



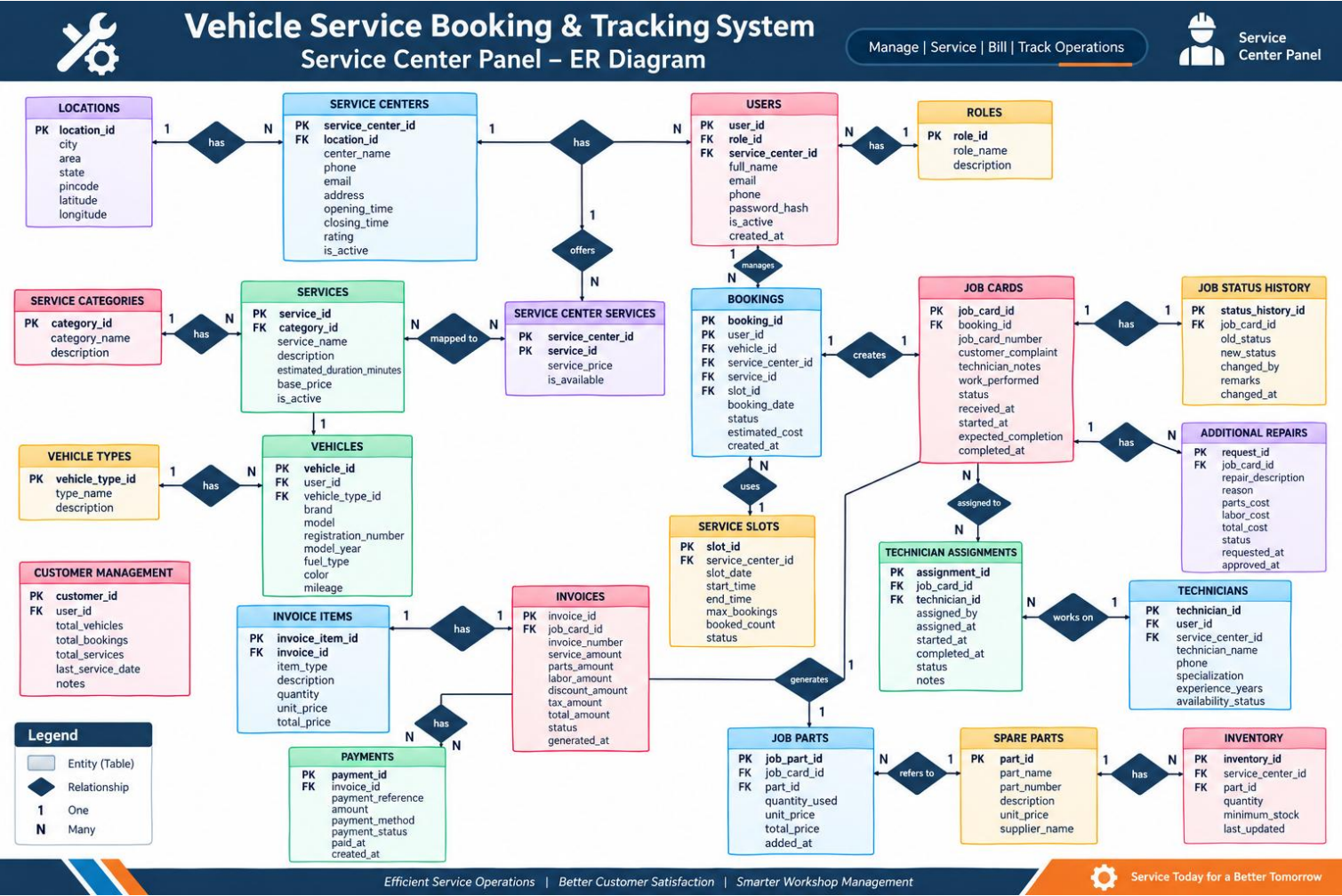
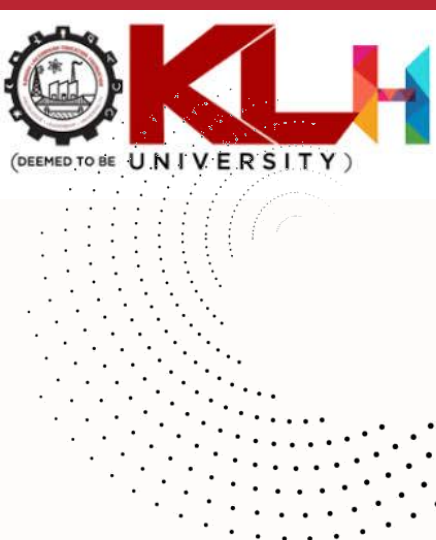
Vehicle Service Booking & Tracking System Customer Panel – ER Diagram

Customer: Book | Track | Pay | Maintain Vehicles



This ER diagram shows the main entities and relationships used in the Customer panel.

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Database Design

Database: MySQL 8.0

Database Name: vehicle_service_db

The database follows a relational database model to store vehicle service information.

Primary Keys (PK) uniquely identify each record.

Foreign Keys (FK) establish relationships between related tables.

The database contains 28 tables covering users, vehicles, service centers, bookings, job cards, technicians, inventory, billing and payments.

Supports the complete service process from service booking → vehicle inspection → service → billing → payment → service history.

Role-based design supports Customer, Service Advisor, Technician and Admin.

Database Tables / Schema

TABLE	PURPOSE
roles	Defines user roles
users	Stores user information
vehicles	Stores customer vehicles
service_centers	Stores service-center details
services	Stores available services
bookings	Stores service bookings
job_cards	Tracks service jobs
technicians	Stores technician details
inventory	Manages spare parts stock
invoices	Stores billing information
payments	Stores payment details
notifications	Stores user notifications

Current Progress

- ✓ **Abstract completed** – Project concept, objectives and overall workflow defined.
- ✓ **Database Design completed** – MySQL database structure, tables, relationships and ER diagrams designed.
- ✓ **Frontend completed** – Customer and Service Center panels with navigation and major interface pages implemented.

Abstract ✓ | Database Design ✓ | Frontend ✓ | Backend  |
Integration  | Testing  **SOON**

Challenges / Issues Faced

- Database Complexity – Designing relationships between users, vehicles, bookings, job cards, inventory and billing.
- Frontend Panel Separation – Creating separate navigation and functionalities for Customer and Service Center users.
- Service Tracking Workflow – Managing multiple service stages from booking to vehicle completion.
- Data Consistency – Maintaining correct relationships between database tables using Primary Keys and Foreign Keys.

Future Work

-  **Backend Development** – REST APIs and backend integration are in progress.
-  **Database Integration** – Connecting the backend with MySQL is in progress.
-  **Testing & Integration** – API testing and frontend-backend integration will be completed next.

Conclusion

- The Vehicle Service Booking & Tracking System provides a centralized solution for managing vehicle servicing.
- It connects customers and service centers through online booking, service tracking, repair approval and billing.
- The system organizes important information such as vehicles, bookings, job cards, technicians, spare parts, invoices, payments and service history.
- The separate Customer and Service Center panels make the system easier to use and manage.
- Overall, the project aims to make vehicle servicing more organized, transparent, efficient and convenient for both customers and service centers.

Thank you